

Appendix: Theoretical Analysis of the Augmented Synthetic Historical Control Estimator (ASHC)

Setup and Assumptions

For the purposes of self-containment, let $\mathcal{T} = \{1, 2, \dots, T\}$ index time at a monthly frequency. Define the pre-treatment period as $\mathcal{T}_1 := \{t \in \mathcal{T} : t \leq T_0\}$ and the post-treatment period as $\mathcal{T}_2 := \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$. Let $m \in \mathbb{N}$ denote the evaluation window length. Define the evaluation period as the final m months of the pre-treatment period $\mathcal{T}_{\text{eval}} := \{T_0 - m + 1, \dots, T_0\} \subset \mathcal{T}_1$.

Let $\mathbf{y} = (y_1, y_2, \dots, y_T)^\top \in \mathbb{R}^T$ denote the observed outcome vector for the treated unit. Define the pre-treatment outcome vector $\mathbf{y}_{\text{pre}} := \mathbf{y}_{\mathcal{T}_1} \in \mathbb{R}^{T_0}$, the evaluation subvector $\mathbf{y}_{\text{eval}} := \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector $\mathbf{y}_{\text{post}} := \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$.

The evaluation subvector $\mathbf{y}_{\text{eval}}$ is the target pattern to be reconstructed using convex combinations of earlier segments from the pre-treatment period.

SHC by [Chen et al. \[2024\]](#) constructs

$$N = T_0 - m - n + 1$$

overlapping temporal segments of length m . Each segment $\tilde{\mathbf{y}}_{[i, i+m-1]}$ is a contiguous subvector of the smoothed pre-treatment series. These comprise the donor matrix

$$\tilde{\mathbf{Y}}_{\text{pre}} := [\tilde{\mathbf{y}}_{[1, m]} \quad \tilde{\mathbf{y}}_{[2, m+1]} \quad \dots \quad \tilde{\mathbf{y}}_{[N, N+m-1]}] \in \mathbb{R}^{m \times N}.$$

Example 0.1. Suppose we have $T_0 = 60$ months of pre-treatment data and we wish to construct donor segments of length $m = 12$ months to predict a post-treatment horizon of $n = 6$ months. The number of eligible segments is:

$$N = 60 - 12 - 6 + 1 = 43.$$

Each donor segment is a vector of 12 consecutive smoothed pre-treatment outcomes.

In the third step, we solve the following program to obtain the optimal weights:

$$\begin{aligned} \mathbf{w}^* = \underset{\mathbf{w} \in \mathbb{R}^N}{\text{argmin}} \quad & \left\| \tilde{\mathbf{y}}_{\text{eval}} - \tilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \\ \text{subject to} \quad & \mathbf{w} \geq \mathbf{0}, \quad \mathbf{1}^\top \mathbf{w} = 1. \end{aligned}$$

Let $\mathbf{w}^{\text{SHC}}$ denote the SHC solution, and $\mathbf{w}^{\text{ASHC}}$ the ASHC solution:

$$\mathbf{w}^{\text{ASHC}} = \arg \min_{\mathbf{w} \in \mathbb{R}^N} \left\{ \frac{1}{2\lambda} \left\| \tilde{\mathbf{y}} - \tilde{\mathbf{Y}} \mathbf{w} \right\|^2 + \left\| \mathbf{w} - \mathbf{w}^{\text{SHC}} \right\|^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2 \right\} \quad \text{s.t.} \quad \sum_{i=1}^N w_i = 1,$$

where $\lambda > 0$ is the regularization parameter, $\varsigma > 0$ is the penalty parameter for the low-variance directions, and $\mathbf{C}_0 \in \mathbb{R}^{N \times k}$ contains the eigenvectors of the Gram matrix $\mathbf{G} = \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}}$ corresponding to eigenvalues less than a threshold $\tau > 0$.

Assumptions

Assumption 0.1 (Linearity with noise). *There exists $\mathbf{w}^* \in \mathbb{R}^N$ such that*

$$\tilde{\mathbf{y}} = \tilde{\mathbf{Y}}\mathbf{w}^* + \boldsymbol{\eta},$$

where $\boldsymbol{\eta} \in \mathbb{R}^m$ is a mean-zero noise vector.

Assumption 0.2 (Bounded noise). $\|\boldsymbol{\eta}\| \leq \delta$ for some $\delta > 0$.

Assumption 0.3 (Smoothness of latent trend). *Let $\ell(t)$ denote the latent smooth component of the treated potential outcome. Suppose $\ell(t)$ is $H + 1$ times differentiable with $|\ell^{(H+1)}(t)| < \epsilon$ for all t and some $H \geq 3$.*

Assumption 0.4 (Operator norm bound). $\|\tilde{\mathbf{Y}}\| \leq \gamma$.

Assumption 0.5 (Simplex feasibility). $\mathbf{w}^{SHC} \in \Delta^{N-1} := \{\mathbf{w} \in \mathbb{R}^N : \sum w_i = 1, w_i \geq 0\}$.

Lemma 0.1 (Deviation from SHC under Affine Constraint). *Let $\Delta := \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$. Under Assumptions (A1)–(A5), we have*

$$\|\Delta\| \leq \frac{\epsilon_{SHC}}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}},$$

where $\epsilon_{SHC} := \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ and $\|\mathbf{C}_0\|^2 := \lambda_{\max}(\mathbf{C}_0^\top \mathbf{C}_0)$.

Proof. We want to bound how much the ASHC weights can deviate from the SHC weights in-sample. To do this, recall that the ASHC solution minimizes

$$J(\mathbf{w}) = \frac{1}{2\lambda} \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{SHC}\|^2 + \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|_2^2 \quad \text{s.t. } \mathbf{1}^\top \mathbf{w} = 1.$$

The first term ensures fidelity to the evaluation vector, the second controls deviation from the SHC weights, and the third stabilizes low-variance directions. The affine constraint $\mathbf{1}^\top \mathbf{w} = 1$ ensures we remain within the feasible weight space.

In order to study the deviation $\Delta := \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$, we require the first-order condition (FOC) of the minimization problem. However, since $J(\mathbf{w})$ is constrained, we cannot obtain the FOC directly by differentiating J . Instead, we introduce a Lagrangian to enforce the affine constraint:

$$\mathcal{L}(\mathbf{w}, \mu) = J(\mathbf{w}) + \mu(\mathbf{1}^\top \mathbf{w} - 1),$$

where $\mu \in \mathbb{R}$ is a multiplier that adjusts the solution so that the weights remain feasible.

Differentiating each component of $\mathcal{L}$ gives

$$\begin{aligned} \nabla_{\mathbf{w}} \frac{1}{2\lambda} \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}\|^2 &= \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top (\tilde{\mathbf{Y}}\mathbf{w} - \tilde{\mathbf{y}}), \\ \nabla_{\mathbf{w}} \|\mathbf{w} - \mathbf{w}^{SHC}\|^2 &= 2(\mathbf{w} - \mathbf{w}^{SHC}), \\ \nabla_{\mathbf{w}} \varsigma \|\mathbf{C}_0^\top \mathbf{w}\|^2 &= 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \mathbf{w}, \\ \nabla_{\mathbf{w}} [\mu(\mathbf{1}^\top \mathbf{w} - 1)] &= \mu \mathbf{1}. \end{aligned}$$

Thus, the stationarity condition $\nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}, \mu) = 0$ becomes

$$\left(\frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top \right) \mathbf{w} = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{y}} + 2\mathbf{w}^{SHC} - \mu \mathbf{1}.$$

Let us denote

$$M := \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma \mathbf{C}_0 \mathbf{C}_0^\top.$$

Subtracting the analogous relation for $\mathbf{w}^{SHC}$ isolates the deviation:

$$M\Delta = \frac{1}{\lambda} \tilde{\mathbf{Y}}^\top (\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{SHC}) - \mu \mathbf{1}.$$

At this point, the Lagrange term appears explicitly. However, the affine constraint requires $\mathbf{1}^\top \Delta = 0$, which means Δ lies in the hyperplane orthogonal to $\mathbf{1}$. Consequently, the term $-\mu \mathbf{1}$ does not affect the component of Δ within the feasible subspace. Geometrically, $\mu \mathbf{1}$ acts along the constraint direction, while Δ is constrained to be orthogonal to it.

Now define the SHC residual $r := \tilde{\mathbf{y}} - \tilde{\mathbf{Y}} \mathbf{w}^{\text{SHC}}$, so that $\|r\| = \epsilon_{\text{SHC}}$. Taking norms, we obtain

$$\|M\Delta\| \leq \frac{1}{\lambda} \|\tilde{\mathbf{Y}}^\top r\| \leq \frac{\gamma}{\lambda} \epsilon_{\text{SHC}},$$

using Assumption (A4). Since the matrix M is positive definite, its smallest eigenvalue is bounded below by $2 + 2\varsigma \|\mathbf{C}_0\|^2$, which ensures strong convexity. Hence

$$\|\Delta\| \leq \frac{\gamma}{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2} \epsilon_{\text{SHC}}.$$

This completes the deviation bound: the difference between ASHC and SHC weights is proportional to the SHC residual, attenuated by the effective regularization strength. $\square$

Remark 0.1. *The key role of the affine constraint $\mathbf{1}^\top \mathbf{w} = 1$ is to introduce a Lagrange multiplier μ , which shifts the optimality condition by a vector in the span of $\mathbf{1}$. Because Δ must satisfy $\mathbf{1}^\top \Delta = 0$, this shift does not affect the conditioning of M and thus does not alter the deviation bound. The effective regularization is governed entirely by the spectrum of M .*

Lemma 0.2. *Under (A4), the error from deviating off SHC weights satisfies*

$$\|\tilde{\mathbf{Y}}\Delta\| \leq \gamma \|\Delta\|.$$

Remark 0.2. *Lemma 0.2 ensures that deviations of the ASHC weights from the SHC weights, measured in ℓ_2 norm, translate into controlled deviations in the predicted outcomes. The operator norm bound (A4) prevents $\tilde{\mathbf{Y}}$ from amplifying small changes in the weights disproportionately.*

Lemma 0.3. *Under (A3), for horizon $k > 0$,*

$$|\ell(T_0 + k) - \ell^{\text{SHC}}(T_0 + k)| \leq b_\epsilon(H, k) := \frac{2\epsilon |k|^{H+1}}{(H+1)!}.$$

Proof. Let $\ell(t)$ denote the latent smooth component of the treated outcome. By (A3), $\ell(t)$ is $(H+1)$ times differentiable with $|\ell^{(H+1)}(t)| \leq \epsilon$.

Step 1. Taylor expansion around T_0 . By Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^H \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with remainder

$$R_{H+1}(k) = \frac{\ell^{(H+1)}(\xi)}{(H+1)!} k^{H+1}, \quad \text{for some } \xi \in [T_0, T_0 + k].$$

Step 2. Bound the remainder.

$$|R_{H+1}(k)| \leq \frac{\epsilon}{(H+1)!} |k|^{H+1}.$$

Step 3. SHC approximation of lower-order terms. By construction, SHC matches the first H terms of this expansion up to error of the same order as the remainder.

Step 4. Combine bounds.

$$|\ell(T_0 + k) - \ell^{\text{SHC}}(T_0 + k)| \leq \frac{2\epsilon}{(H+1)!} |k|^{H+1}.$$

$\square$

Remark 0.3. Lemma 0.3 formalizes the SHC bias as a Taylor remainder. The SHC approximation matches derivatives of $\ell(t)$ at T_0 up to order H , leaving bias decaying like $|k|^{H+1}$. Thus, for smooth latent trends and moderate horizons, SHC bias is negligible.

Main Result

Proposition 0.1. Under assumptions (A1)–(A5), the ASHC prediction error is bounded by

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}} \right) + b_\epsilon(H, k) + \delta.$$

Proof. We want to bound the prediction error of the ASHC estimator,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\|.$$

Step 1: Decompose the error using the triangle inequality. Let $\mathbf{w}^*$ denote the ideal weights from the SHC estimator without the ASHC penalty. Define the difference $\Delta := \mathbf{w}^{\text{ASHC}} - \mathbf{w}^*$. Then,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| = \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^* - \tilde{\mathbf{Y}}\Delta\| \leq \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^*\| + \|\tilde{\mathbf{Y}}\Delta\|.$$

The first term, $\epsilon_{\text{SHC}} := \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^*\|$, is the SHC prediction error without augmentation.

Step 2: Control the second term using operator norm. By Lemma 0.2, the matrix $\tilde{\mathbf{Y}}$ satisfies

$$\|\tilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|,$$

where γ is an operator norm bound capturing how the weights perturbations translate into prediction changes.

Step 3: Bound the weight difference $\|\Delta\|$. From Lemma 1 (stated earlier), the ASHC weight vector $\mathbf{w}^{\text{ASHC}}$ stays close to the SHC weights in Euclidean norm:

$$\|\Delta\| = \|\mathbf{w}^{\text{ASHC}} - \mathbf{w}^*\| \leq \frac{\epsilon_{\text{SHC}}}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}},$$

where the denominator reflects the strength of the ASHC penalty, ensuring stability.

Step 4: Incorporate smoothness bias and noise. Finally, the total prediction error also includes the smoothness bias $b_\epsilon(H, k)$ due to approximating the latent function by a truncated Taylor expansion of order H , and a noise term bounded by δ (from assumption (A2)).

Putting all these together, we get

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}\mathbf{w}^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} + \gamma\|\Delta\| + b_\epsilon(H, k) + \delta \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}} \right) + b_\epsilon(H, k) + \delta,$$

which completes the proof. $\square$

Discussion

Proposition~0.1 decomposes ASHC error into: 1. SHC fit error (ϵ_{SHC}), 2. amplification from ASHC deviation, 3. smoothness bias $b_\epsilon(H, k)$, 4. noise δ .

The penalty term $\varsigma\|\mathbf{C}_0^\top \mathbf{w}\|_2^2$ reduces deviation but may bias if $\mathbf{w}^*$ has components in $\mathbf{C}_0$ directions. The trade-off between λ and ς governs stability vs. fidelity.

Asymptotic Results

Theorem 0.1. Under (A1)–(A5), as $m \rightarrow \infty$ and $H \rightarrow \infty$, for fixed $k > 0$,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{ASHC}\| \rightarrow \delta.$$

If $\boldsymbol{\eta}$ consists of independent mean-zero sub-Gaussian entries, then

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{ASHC}\| \xrightarrow{p} 0.$$

Proof. Starting from Proposition 0.1, we have

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{ASHC}\| \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}} \right) + b_\epsilon(H, k) + \delta.$$

Step 1. Behavior of ϵ_{SHC} . By (A3), $\ell(t)$ is $(H+1)$ -times differentiable. Gaussian kernel smoothing in SHC ensures

$$\epsilon_{\text{SHC}} \rightarrow 0 \quad \text{as } m \rightarrow \infty, H \rightarrow \infty.$$

Chen et al. [2024]

Step 2. Behavior of ASHC penalty term. Since $\epsilon_{\text{SHC}} \rightarrow 0$, the inflation factor

$$\frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}} \epsilon_{\text{SHC}}$$

vanishes asymptotically.

Step 3. Behavior of smoothness bias $b_\epsilon(H, k)$. From Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^H \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with

$$|R_{H+1}(k)| \leq \frac{\epsilon}{(H+1)!} |k|^{H+1}.$$

Thus

$$b_\epsilon(H, k) \leq \frac{2\epsilon}{(H+1)!} |k|^{H+1} \rightarrow 0 \quad \text{as } H \rightarrow \infty.$$

Step 4. Behavior of noise term. By (A2), $\|\boldsymbol{\eta}\| \leq \delta$. If $\boldsymbol{\eta}$ is i.i.d. sub-Gaussian with variance proxy σ^2 , then

$$\frac{\|\boldsymbol{\eta}\|}{m} \xrightarrow{p} 0.$$

Step 5. Combine bounds. Therefore,

$$\limsup_{m \rightarrow \infty} \|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{ASHC}\| \leq \delta.$$

With sub-Gaussian noise, the limit is 0 in probability. $\square$

Corollary 0.1. Under (A1)–(A5), the ASHC estimator is asymptotically unbiased up to δ . If $\boldsymbol{\eta}$ is i.i.d. mean-zero sub-Gaussian with variance proxy σ^2 and bandwidth $h_m \asymp m^{-1/(2H+3)}$, then as $m \rightarrow \infty$,

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{ASHC}\| = O_p\left(m^{-H/(2H+3)}\right).$$

For large H , this approaches the parametric rate $O_p(m^{-1/2})$.

Proof. Part 1: Consistency. From Theorem 0.1, the error converges to δ , or 0 in probability under sub-Gaussian noise.

Part 2: Convergence Rate. From Proposition 0.1, we have

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{\text{ASHC}}\| \leq \epsilon_{\text{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}} \right) + b_\epsilon(H, k) + \delta.$$

Step 1. Bias term. From Lemma 0.3,

$$b_\epsilon(H, k) = O(h_m^H).$$

Step 2. Stochastic error term. With sub-Gaussian noise, the smoothed variance is

$$O_p((mh_m)^{-1/2}).$$

Step 3. SHC fit error. Combining,

$$\epsilon_{\text{SHC}} = O_p(h_m^H + (mh_m)^{-1/2}).$$

Step 4. Optimal bandwidth. Equating orders $h_m^H \asymp (mh_m)^{-1/2}$ yields

$$h_m \asymp m^{-1/(2H+3)}.$$

Step 5. Convergence rate. Substituting,

$$\epsilon_{\text{SHC}} = O_p(m^{-H/(2H+3)}).$$

Thus the total error is

$$\|\tilde{\mathbf{y}} - \tilde{\mathbf{Y}}w^{\text{ASHC}}\| = O_p(m^{-H/(2H+3)}).$$

Step 6. Near-parametric rate. As $H \rightarrow \infty$,

$$\frac{H}{2H+3} \rightarrow \frac{1}{2}.$$

So the rate approaches $O_p(m^{-1/2})$. □

Remark 0.4. The deterministic noise bound δ in (A2) corresponds to $\|\boldsymbol{\eta}\| \leq \delta$. Under sub-Gaussian noise, $\delta = O(\sigma\sqrt{m})$, while kernel smoothing reduces effective noise to $O_p((mh_m)^{-1/2})$. This explains the appearance of the nonparametric rate.

References

Yi-Ting Chen, Jui-Chung Yang, and Tzu-Ting Yang. Synthetic historical control for policy evaluation. Available at SSRN: <https://ssrn.com/abstract=4995085>, September 2024. URL <http://dx.doi.org/10.2139/ssrn.4995085>.